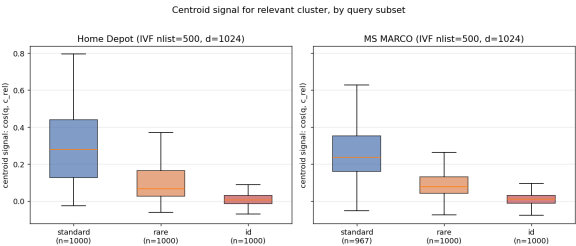


Direct evidence: measure the centroid signal

For each query: compute $\langle q, c_{\text{rel}} \rangle$ on the centroid of the query’s *relevant* cluster (IVF, $n_{\text{list}}=500$, $d=1024$, HD).



The bound $\langle c, q \rangle \approx \alpha_{t^*}^2 p$ is not just suggestive — it is what we *measure*.

Beyond Representation: Index Structure Limitations in Dense Retrieval

Direct evidence: measure the centroid signal



Framing. This is where we stop arguing from theory and start measuring the exact quantity the bound predicts.

What we did (per query). Train IVF on Home Depot ($n_{\text{list}} = 500$, $d = 1024$). For every query, look up which cluster actually contains its relevant doc(s) — call that centroid c_{rel} . Then compute $\langle q, c_{\text{rel}} \rangle$, and also *where* c_{rel} ranks among all 500 centroids scored against q , and whether it makes the top 250 (“hit@250” — the halfway mark).

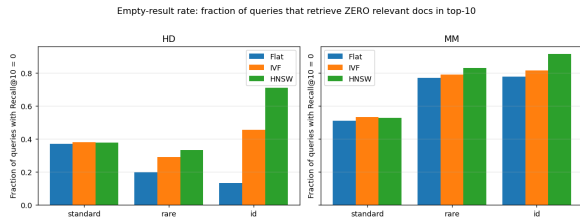
Histogram. Standard queries: signal ~ 0.28 , clearly separated from zero. Rare queries collapse toward zero (~ 0.07). ID queries pile up at ~ 0.007 — **40× smaller than standard**.

Table. Standard: correct centroid at rank 2 on average, hit@250 = 97% — IVF with $n_{\text{probe}} = 5$ is basically always correct. Rare: rank 69, so you’d need $n_{\text{probe}} \approx 70$ (huge). **ID: rank 216 out of 500 — worse than median.** hit@250 = 59% means **41% of ID queries have their relevant cluster in the bottom half** of the centroid ranking. Those are structurally unrecoverable short of scanning all clusters — which defeats IVF’s purpose.

Tie to the bound. $\langle c, q \rangle \approx \alpha_{t^*}^2 p$. Rare tokens: huge α (IDF), tiny p (prevalence). Product falls under the JL floor ϵ — exactly the collapse we measure.

Anticipated question. “Just use finer clusters?” It helps (rare/ID gain 2–3× more relatively than standard going $n_{\text{list}} 200 \rightarrow 1000$) but you’re fighting p : shrinking clusters means more centroids to score,

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Fraction of queries with zero relevant docs in top-10:

- ▶ HD-id: Flat **13%** → HNSW **71%**
- ▶ MM-id: Flat **78%** → HNSW **92%**
- ▶ Standard: within ± 2 pp on both datasets.

Cost to match Flat recall (efSearch, HD):

- ▶ standard: **$2.7\times$ faster** than Flat
- ▶ id: **$40\times$ slower** than Flat

ANN's speedup evaporates on the population that needs it most.